CURRICULUM VITÆ

1 Personal information

Name: Junior Rodrigues Ribeiro
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2 Professional summary

PhD in Computational Mathematics with strong experience in mathematical modeling and numerical simulation, specializing in the control of dynamic systems (risk modeling and control in variable scenarios). Possess practical expertise in the three topics of data science:

- **Data Analysis**: Statistics, trend analysis, data visualization, and modeling.
- **Optimization (Operations Research)**: Applying models to optimize decisions, aiming to minimize costs/risks or maximize efficiency and profit.
- **Machine Learning (Neural Networks)**: Developing advanced systems for forecasting, interpretation, and data classification.

Proficient in the technical data stack, including Python, MATLAB, SQL, and Microsoft Power BI. My strong analytical ability allows me to translate complex theoretical knowledge into practical solutions for business challenges.

Seeking a Data Science, Analysis or correlated position where I can apply my modeling and optimization skills to generate measurable results and add strategic value to the company.

3 Academic background

- Ph.D. in Computational Mathematics (2019–2024)
  - Institution: Institute of Mathematical and Computer Sciences (ICMC-USP), São Carlos–SP.
  - Thesis: "Stochastic linear systems with jumps in the parameters and in the time domain".
  - Research Focus: Developed mathematical models for stability analysis and risk control in dynamic systems subject to abrupt scenario changes.
- M.S. in Computational Mathematics (2016–2019)
  - Institution: Institute of Mathematical and Computer Sciences (ICMC-USP), São Carlos–SP.
  - Dissertation: "Analysis and improvement of the variational method for control of Markov jump linear systems with no jump observation".
  - Research Focus: Developed numerical optimization strategies for control problems in scenarios involving partial data or incomplete observation.
- B.S. in Mathematics (2012–2015)
  - Institution: State University of Mato Grosso do Sul (UEMS), Dourados–MS.

4 Professional experience

Substitute Mathematics Professor (August 2024 – July 2025)
Federal University of Mato Grosso do Sul (UFMS)
Responsible for teaching mathematics subjects for various undergraduate courses.

5 Academic research and publications

Author of scientific articles in the field of Markov Jump Linear Systems (MJLS), with publications and works in progress. Three most relevant articles:

- L. H. Romero, J. R. Ribeiro and E. F. Costa, *"On the H2 Control of Hidden Markov Jump Linear Systems,"* in IEEE Control Systems Letters, vol. 7, pp. 1315-1320, 2023, DOI: 10.1109/LCSYS.2023.3236892. [Cited 7 times]
- Lozada-Cruz, G., Rubio-Mercedes, C. E., Rodrigues-Ribeiro, J. (2018). *"Numerical Solution of Heat Equation with Singular Robin Boundary Condition"*. Trends in Computational and Applied Mathematics, 19(2), 209. DOI: 10.5540/tema.2018.019.02.209. [Cited 5 times]
- J. R. Ribeiro, L. H. Romero, E. F. Costa and M. G. Todorov, *"Comments on the H2 Norm of Discrete-Time Stochastic Jump Parameter Linear Systems,"* in IEEE Control Systems Letters, vol. 7, pp. 1470-1475, 2023, DOI: 10.1109/LCSYS.2023.3268018. [Cited once]

6 Applied research

Co-author of the scientific report for a study by Carteira Global[1]

- *"Portfolio Selection with minimum investment constraints"*. **Carteira Global**. VII Workshop de Soluções Matemáticas para Problemas Industriais. CeMEAI-USP, 2021. Disponível em https://cemeai.icmc.usp.br/WSMPI/7a-edicao, access on 2025, September 11^{th} at 00:34h.

  In this study, our team worked on a portfolio optimization problem to select the best portfolio[2] of assets given user inputs (choice of expected risk and return – Sharpe index).

Co-author of the scientific report for a study by B2W Digital[3] (now Americanas S.A.)

- "Roteirização dinâmica". **B2W Digital**. IV Workshop de Soluções Matemáticas para Problemas Industriais. CeMEAI-USP, 2018. Disponível em https://www.cemeai.icmc.usp.br/4WSMPI, access on 2025, September 10^{th} at 23:24h.

  In this study, our team tackled the problem of calculating routes for goods deliveries using real data provided by the company. We created a mathematical model of the problem and submitted it to a solver to calculate a solution (a set of routes) that would minimize the sum of costs: fixed costs, delivery person bonuses, work hours, fleet rental, and distances traveled.

[1] Instagram of Carteira Global https://www.instagram.com/carteiraglobal.
[2] Description of the problem https://youtu.be/zGHCKNZynsc (video in Portuguese).
[3] LinkedIn of B2W Digital https://www.linkedin.com/company/b2w—companhia-global-do-varejo.

7 Skills and abilities

1. Languages and Databases
   - Languages: Python (Proficient), SQL (Proficient), MATLAB;
   - Databases: Experience with Relational Databases (SQL);
   - Version Control: Git e GitHub.
2. Data Analysis and Business Intelligence
   - BI Tools: Microsoft Power BI;
   - Analysis and Computation: Pandas, NumPy, Matplotlib;
   - Quantitative Analysis: Mathematical Modeling, Applied Probability and Statistics (Regression).
3. Machine Learning (ML) and Optimization
   - ML Libraries: Scikit-Learn, Keras;
   - Techniques: Optimization (Operation Research), Dynamic Systems Control, Numerical Simulation;
   - Data Processing: Multiprocessing, BeautifulSoup.
4. Languages
   - Brazilian Portuguese: native.
   - English: Intermediate (Focus on Technical Reading and Writing);
   - Spanish: Beginner (Comprehension and Speaking).
5. Soft Skills
   - Analytical Capacity and Complex Problem-Solving;
   - Self-Learning, Proactivity and Adaptability;
   - Technical Communication (Oral and Written);
   - Critical Thinking and Emotional Intelligence.